

Final Year Project Report

Hotel & Restaurant Management System (HRMS)

Project Title	Hotel & Restaurant Management System (HRMS)
Submitted By	_____ (Student Name)
Enrollment No.	_____
Course / Branch	B.E. / B.Tech (Computer / IT)
Semester	8th Semester
Submitted To	_____ (Guide Name)
Department	Computer Engineering / Information Technology
College	_____ (College Name)
Academic Year	2025-26

Declaration: I hereby declare that this project report is my original work carried out under the guidance of my project guide.

Signature of Student: _____

Signature of Guide: _____

Certificate

This is to certify that the project titled “**Hotel & Restaurant Management System (HRMS)**” is a bonafide work carried out by _____ (Enrollment No. _____) in partial fulfillment of the requirements for the award of the degree of **B.E./B.Tech in Computer Engineering / Information Technology** during the academic year **2025-26**.

The work has been carried out under my supervision and to the best of my knowledge, the project work is original and has not been submitted elsewhere.

Project Guide: _____

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Principal: _____

Acknowledgement

I would like to express my sincere gratitude to my project guide _____ for continuous support, guidance and motivation throughout the project work. I am thankful to the Head of Department and all faculty members for their valuable suggestions and encouragement.

I also thank my friends and family for their constant support and

cooperation. Student Name: _____

Abstract

The Hotel & Restaurant Management System (HRMS) is a comprehensive web-based application developed to automate and integrate the core operational activities of hotels and restaurants within a single unified platform. In the hospitality industry, managing room reservations, customer details, restaurant orders, billing, and reporting manually or through fragmented systems often leads to inefficiencies, human errors, data inconsistency, and poor customer experience. The HRMS project aims to address these challenges by providing a centralized, secure, and scalable solution that simplifies daily operations and enhances service quality.

The system supports role-based access control, enabling different users such as Admin, Receptionist, and Restaurant Staff to perform their respective tasks efficiently. The Admin module allows complete control over system configuration, user management, room details, menu items, pricing, and reports. The Receptionist module focuses on hotel-related operations including customer registration, room availability tracking, booking management, and check-in/check-out processes. The Restaurant Staff module manages menu categories, food items, table allocation, order processing, and order status tracking. This structured division of responsibilities ensures operational clarity and data security.

HRMS provides end-to-end automation of the booking lifecycle, starting from room availability checks to customer check-in, stay management, and check-out with automatic bill calculation. On the restaurant side, the system manages menu availability, order placement, order tracking, and billing, enabling faster service and accurate invoicing. A unified billing system generates invoices that may include both hotel accommodation and restaurant charges, reducing manual effort and billing discrepancies.

The application is developed using modern web technologies such as Node.js and Express.js for backend logic, EJS templating for dynamic user interfaces, and MongoDB as a NoSQL database for flexible and efficient data storage. The front-end is designed using HTML, CSS, and JavaScript to provide a responsive and user-friendly interface accessible across devices. The system follows an MVC-based architecture, which improves maintainability, scalability, and code organization.

In addition to operational features, HRMS includes reporting and dashboard functionalities that provide insights into room occupancy, daily sales, and popular menu items, supporting informed decision-making for management. Overall, the Hotel & Restaurant Management System enhances operational efficiency, minimizes manual errors, improves customer satisfaction, and provides a reliable digital solution suitable for small-to-medium hospitality businesses.

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1. Introduction

The hospitality industry is one of the fastest-growing service sectors, where efficiency, accuracy, and customer satisfaction play a crucial role in overall business success. Hotels and restaurants are required to manage multiple operations simultaneously, such as room reservations, customer records, restaurant orders, billing, and reporting. Traditionally, many small and medium-scale hotels and restaurants rely on manual processes or disconnected software systems, which often result in data redundancy, operational delays, billing errors, and poor coordination between departments. With the increasing demand for digital transformation, there is a strong need for an integrated and automated management system.

The Hotel & Restaurant Management System (HRMS) is designed to address these challenges by providing a centralized web-based platform that efficiently manages both hotel and restaurant operations. The system integrates room booking, customer management, restaurant menu handling, order processing, billing, and reporting into a single application. By automating routine tasks, HRMS minimizes human intervention, reduces errors, and improves overall operational efficiency.

In hotel operations, managing room availability, check-in and check-out processes, and customer details is critical. Manual handling of these tasks can lead to problems such as double bookings, incorrect billing, and loss of customer data. HRMS provides a structured approach to room management, enabling real-time tracking of room status, automated booking workflows, and accurate calculation of stay charges. This ensures smooth front-desk operations and enhances the guest experience.

Similarly, restaurant operations involve menu management, table allocation, order tracking, and payment processing. During peak hours, manual order handling can cause delays, miscommunication, and billing mistakes. The HRMS restaurant module streamlines these activities by maintaining digital menus, managing orders efficiently, and tracking order status from placement to payment. This improves coordination between staff and ensures timely service delivery.

The system is built using modern web technologies such as Node.js and Express.js for server-side development, EJS for dynamic user interface rendering, and MongoDB for flexible and scalable data storage. The application follows a Model-View-Controller (MVC) architecture, which enhances code organization, maintainability, and scalability. Role-based access control ensures that only authorized users can access specific system functionalities, improving security and data integrity.

Overall, the Hotel & Restaurant Management System aims to provide a reliable, user-friendly, and cost-effective solution for hospitality businesses. By integrating hotel and restaurant operations into a single platform, HRMS helps organizations improve efficiency, enhance customer satisfaction, and make better management decisions through accurate data and reports.

2. Problem Statement

In the hospitality industry, effective management of hotel and restaurant operations is essential to ensure smooth service delivery and customer satisfaction. However, many small and medium-sized hotels and restaurants still rely on manual processes or partially automated systems to manage room bookings, customer information, restaurant orders, and billing. These traditional approaches are time-consuming, error-prone, and inefficient, especially during peak business hours. As a result, organizations often face operational challenges that negatively impact service quality and business performance.

One of the major problems in hotel management is the lack of a centralized system for handling room availability, reservations, and check-in/check-out processes. Manual record keeping can lead to issues such as double booking of rooms, incorrect room status updates, delayed check-ins, and inaccurate billing. Maintaining customer records on paper or in separate files also increases the risk of data loss, duplication, and security breaches. These problems make it difficult for hotel staff to provide fast and reliable service to guests.

Similarly, restaurant management faces challenges in menu handling, order processing, and billing. Manual order taking and billing can cause miscommunication between staff, delayed order preparation, and calculation errors in invoices. During busy hours, tracking table occupancy and order status becomes difficult, leading to poor coordination and customer dissatisfaction. The absence of an integrated order management system also makes it hard for management to analyze sales performance and popular menu items.

Another significant issue is the lack of integration between hotel and restaurant operations. In many establishments, hotel accommodation charges and restaurant bills are managed separately, resulting in inconsistent data and additional workload during billing. This separation prevents the generation of unified invoices and comprehensive reports, making financial tracking and decision-making more complex.

Furthermore, existing systems often lack proper access control and reporting features. Without role-based access, sensitive data may be exposed to unauthorized users. The absence of real-time reports on occupancy, sales, and revenue limits management's ability to monitor performance and plan future improvements.

Therefore, there is a clear need for a secure, integrated, and automated Hotel & Restaurant Management System that can efficiently manage hotel and restaurant operations within a single platform. Such a system should reduce manual effort, minimize errors, improve coordination among departments, and provide accurate reporting to support effective decision-making.

3. Objectives

The primary objective of the Hotel & Restaurant Management System (HRMS) is to develop an integrated, web-based application that automates and streamlines hotel and restaurant operations to improve efficiency, accuracy, and customer satisfaction. The system aims to replace manual and fragmented processes with a centralized digital platform that supports smooth coordination between different departments of a hospitality organization.

One of the key objectives of this project is to provide a centralized management system for handling hotel room bookings and customer information. The system aims to enable real-time tracking of room availability, efficient management of reservations, and smooth execution of check-in and check-out processes. By automating these tasks, the system reduces the chances of double bookings, incorrect room allocation, and billing errors, thereby improving the overall guest experience.

Another important objective is to automate restaurant operations, including menu management, table allocation, order processing, and billing. The system is designed to ensure accurate order handling, timely updates of order status, and efficient coordination among restaurant staff. This helps reduce service delays, minimize order-related errors, and enhance customer satisfaction during peak business hours.

The project also aims to implement role-based access control to ensure system security and operational clarity. Different user roles such as Admin, Receptionist, and Restaurant Staff are provided with access to only those functionalities relevant to their responsibilities. This objective enhances data security, prevents unauthorized access, and ensures accountability in system usage.

An additional objective of HRMS is to provide an integrated billing system that can generate accurate invoices for hotel accommodation, restaurant orders, or combined services. Automated bill calculation, including taxes and service charges, reduces manual effort and ensures transparency and correctness in financial transactions.

The system also aims to generate meaningful reports and dashboards to support management decision-making. Reports related to room occupancy, daily sales, revenue trends, and popular menu items help management analyze performance and plan future improvements.

Finally, the project aims to develop a user-friendly, scalable, and cost-effective solution using modern web technologies such as Node.js, Express.js, EJS, and MongoDB. The system is designed to be easily maintainable and extensible, allowing future enhancements such as online payment integration, advanced analytics, and multi-branch support.

4. Scope of the Project

The scope of the Hotel & Restaurant Management System (HRMS) defines the functional boundaries and capabilities of the proposed system. The project is focused on developing a web-based application that integrates hotel and restaurant operations into a single, centralized platform. The system is intended to support the day-to-day operational needs of small to medium-sized hotels and restaurants by automating core processes and reducing manual workload.

Within the hotel management domain, the scope of the project includes room management, customer management, and booking lifecycle management. The system allows authorized users to add, update, and manage room details such as room type, pricing, and availability status. Customer information including personal details and identification data can be securely stored and managed. The booking process covers reservation creation, room allocation, check-in, check-out, and automatic calculation of stay charges based on duration and room pricing. Real-time room status updates ensure accurate availability tracking and prevent double bookings.

In the restaurant management domain, the scope includes menu management, table management, order processing, and billing. The system supports the creation and maintenance of menu categories and items along with pricing and availability status. Table allocation and occupancy tracking are included to manage dine-in operations efficiently. Restaurant staff can create, update, and track orders, ensuring smooth communication and timely service. Automated calculation of order totals reduces billing errors and improves service accuracy.

The project also covers an integrated billing and invoicing system that supports hotel services, restaurant services, or combined invoices. The system automatically applies applicable taxes and service charges, providing accurate and transparent billing. Generated invoices can be stored for record-keeping and future reference.

Another important aspect within the scope is role-based access control and system security. Different user roles such as Admin, Receptionist, and Restaurant Staff are provided with controlled access to system functionalities based on their responsibilities. This ensures data security, accountability, and organized system usage.

The scope further includes basic reporting and analytical features that provide insights into hotel occupancy, restaurant sales, and overall revenue. These reports support managerial decision-making and operational planning.

The project scope is limited to internal management operations and does not include advanced features such as online booking portals for customers, online payment gateways, mobile applications, or multi-branch management. However, the system is designed in a scalable manner, allowing such features to be added in future enhancements.

5. Proposed System

The proposed system is a **web-based Hotel & Restaurant Management System (HRMS)** developed to automate and manage the complete hotel booking process along with restaurant operations in a single integrated platform. The main purpose of this system is to replace the traditional manual and paper-based working method with a **digital, fast, accurate, and secure management solution**. In many hotels, room booking, customer details, restaurant orders, billing, and reporting are handled manually which increases the chances of errors, delays, and data loss. To solve these issues, this HRMS provides a centralized application where all hotel and restaurant activities can be performed smoothly through a browser.

This proposed system is designed using **Node.js and Express.js** for server-side development, **MongoDB** for database management, and **EJS templates with HTML, CSS, and JavaScript** for creating a user-friendly interface. The system follows an organized structure (MVC pattern) where data is stored securely, operations are controlled properly, and UI pages are simple for staff to understand. The proposed system supports **role-based login**, which means different staff members will access different modules according to their responsibility. For example, **Admin** can manage complete system settings and staff accounts, **Receptionist** can manage customer records and hotel bookings, and **Restaurant Staff** can handle menu and order management. This role-based approach ensures better security and prevents unauthorized access.

The proposed HRMS provides a complete hotel management workflow such as **room management, customer registration, room booking, check-in/check-out, and status tracking**. Rooms can be marked as available, occupied, or under maintenance, and the receptionist can easily allocate rooms without confusion or double-booking. Booking records are stored with customer details, check-in date, check-out date, and total stay charges. The system automatically calculates the total bill based on duration of stay and room price, which reduces calculation mistakes and saves time.

Along with hotel operations, the proposed system also includes restaurant features like **menu category management, menu item management, table management, and order processing**. Restaurant staff can create dine-in or takeaway orders, select items from the menu, update quantities, and track order status such as pending, cooking, served, and paid. This makes restaurant workflow more organized and improves customer service speed. The system also helps to manage item availability and pricing updates easily without rewriting menus manually.

A major advantage of the proposed system is its **billing and invoice generation module**. The system generates invoices for hotel bookings as well as restaurant orders. It supports combined billing if a customer uses both hotel and restaurant services. Taxes, service charges, and total payable amount are calculated automatically. This ensures transparency in billing and reduces manual mistakes. The invoice records are stored in the database for future reference and reporting purposes.

The proposed HRMS also provides **searching, filtering, and reporting features** to help management take better decisions. Admin can view daily sales, occupancy reports, total bookings, popular menu items, and revenue summaries. These reports help the hotel owner or manager to analyze performance, manage staff efficiency, and plan improvements. Since the system is digital, all data is stored securely in MongoDB with backup possibilities, which prevents loss of important customer and business records.

Overall, the proposed Hotel & Restaurant Management System (HRMS) offers a complete digital solution that improves hotel and restaurant operations by making them **faster, reliable, secure, and well-organized**. It reduces paperwork, improves staff productivity, enhances customer satisfaction, and provides better control over bookings, orders, billing, and reporting. This system is suitable for small to medium hotels and restaurants where automation is required to improve efficiency and service quality.

6. Feasibility Study

Feasibility Study is an important step in software development that helps to analyze whether the proposed system is practical, beneficial, and possible to implement successfully. It ensures that the project can be completed within the available resources such as time, cost, technology, and manpower. For the **Hotel & Restaurant Management System (HRMS)**, the feasibility study is performed to check the project's success possibility and to confirm that it can fulfill hotel and restaurant operational requirements efficiently. The feasibility study for this project mainly includes **Technical Feasibility, Economic Feasibility, Operational Feasibility, and Schedule Feasibility**.

1) Technical Feasibility

Technical feasibility checks whether the required technologies, tools, and technical skills are available to develop and run the system successfully. The proposed HRMS is built using modern and widely used web technologies such as **Node.js, Express.js, MongoDB, and EJS templates** with HTML, CSS, and JavaScript. These technologies are open-source, easy to learn, and supported by a large developer community. Node.js provides fast server-side execution and supports asynchronous operations, which makes it suitable for handling multiple hotel and restaurant requests at the same time. MongoDB is used as a database because it is flexible, scalable, and supports quick storage and retrieval of booking and order information. The required software tools like **VS Code, Git, and MongoDB Compass** are easily available. Since the system is web-based, it can run on any device with an internet browser, making it technically feasible for real-world usage. Therefore, the project is technically feasible because the development technologies are available and suitable for building this system.

2) Economic Feasibility

Economic feasibility determines whether the system is cost-effective and whether the organization can afford the project development and maintenance cost. This HRMS project is economically feasible because it uses **free and open-source technologies** like Node.js, Express.js, MongoDB, and EJS, which reduces software licensing costs. The system does not require expensive hardware because it can run on a basic computer system with minimum requirements. For deployment, it can be hosted on affordable cloud platforms or even locally in the hotel environment. The major investment required is only in terms of development time and basic infrastructure. In return, the system helps to reduce manual work, paperwork, and human errors, which saves operational cost. It also improves efficiency in booking, billing, and restaurant order management, leading to better customer satisfaction and increased business growth. Hence, the system is economically feasible and beneficial for small to medium hotels and restaurants.

3) Operational Feasibility

Operational feasibility checks whether the system can be used effectively by the staff members and whether it will be accepted in daily operations. The proposed HRMS is designed with a simple and user-friendly interface so that staff members like receptionists, restaurant staff, and admin can easily use it with minimum training. The system provides role-based access, which means each user can access only their required modules, making the workflow easy and secure. Receptionists can manage bookings, check-in/check-out, and customer details, while restaurant staff can manage menu items and orders. The system reduces manual workload and speeds up operations, which improves staff productivity. Since all data is stored digitally, it becomes easy to track booking history, customer information, restaurant sales, and invoices. Therefore, this project is operationally feasible because it improves daily hotel and restaurant operations and can be adopted smoothly by users.

4) Schedule Feasibility

Schedule feasibility evaluates whether the project can be completed within the planned time period. The HRMS project is manageable and can be completed within the academic project timeline because it is divided into multiple modules such as authentication, room management, booking management, menu management, order management, billing, and reports. Each module can be developed step-by-step and tested individually. Since the tools and frameworks used in this project provide ready-made libraries and fast development support, it reduces the overall development time. Proper planning and implementation in phases make the project achievable within the given schedule. Hence, the project is schedule feasible.

7. System Requirements

System Requirements define the necessary hardware, software, and other resources required to develop and run the proposed application successfully. The **Hotel & Restaurant Management System (HRMS)** is a web-based application designed to manage hotel bookings and restaurant operations efficiently. Since it is developed using modern web technologies, the system can run smoothly on standard computers and does not require very high-end hardware. The requirements are categorized into **Hardware Requirements, Software Requirements, and Other Supporting Requirements** to ensure proper installation, development, and execution of the project.

1) Hardware Requirements

The hardware requirements specify the minimum and recommended computer specifications needed for developing and running the HRMS system. The application can be executed on a single system for local usage or can be hosted on a server for multiple users.

Minimum Hardware Requirements

- **Processor:** Dual-Core Processor (Intel i3 or equivalent)
- **RAM:** 4 GB
- **Hard Disk / Storage:** 10 GB free space
- **Display:** 14-inch monitor or above
- **Internet Connection:** Basic broadband connection (for cloud deployment and updates)
- **Input Devices:** Keyboard and Mouse

Recommended Hardware Requirements

- **Processor:** Quad-Core Processor (Intel i5/i7 or equivalent)
- **RAM:** 8 GB or more
- **Storage:** 20 GB or more free space (SSD recommended for faster performance)
- **Internet Connection:** Stable broadband for multi-user access and cloud database usage

The above hardware setup is sufficient for hotel staff like receptionists, admin, and restaurant staff to use the system without performance issues.

2) Software Requirements

Software requirements define the operating system, programming tools, frameworks, database, and browser required to develop and run the HRMS system.

Operating System

The system can be developed and executed on any modern operating system such as:

- **Windows 10 / 11**
- **Linux (Ubuntu or similar)**
- **macOS (optional)**

Development Tools

To build and manage the project, the following tools are required:

- **Visual Studio Code (VS Code):** For writing and managing the project code
- **Git / GitHub:** For version control and project backup
- **Postman:** For testing API routes (optional but useful)
- **MongoDB Compass:** For viewing and managing MongoDB database (optional)

Backend Requirements

The backend is responsible for handling server operations, business logic, and database connectivity:

- **Node.js (LTS Version):** Server-side runtime environment
- **Express.js:** Framework for routing and server management
- **Mongoose:** ODM library to connect Node.js with MongoDB database

Frontend Requirements

The frontend provides a user-friendly interface for staff to use the system:

- **HTML5:** For page structure
- **CSS3:** For styling and layout
- **JavaScript:** For dynamic behavior and validations
- **EJS (Embedded JavaScript Templates):** For server-side rendering and dynamic UI pages

Database Requirements

The system stores all hotel and restaurant information securely in:

- **MongoDB:** NoSQL database for storing users, rooms, bookings, orders, invoices, and reports

Browser Requirements

Since HRMS is web-based, it requires a browser for execution:

- **Google Chrome (Recommended)**
- **Microsoft Edge**
- **Mozilla Firefox**
- **Safari (optional)**

3) Security and Additional Requirements

For secure and smooth working of the system, some additional requirements are also necessary:

- **Authentication System:** Secure login for Admin, Receptionist, and Restaurant Staff
- **Password Encryption:** Using bcrypt hashing for password security
- **Session Management:** To maintain logged-in user state securely
- **Role-Based Access Control:** Different permissions for different staff roles
- **Backup Facility:** Regular backup of database to prevent data loss
- **Error Handling:** Proper validation and error messages to avoid system crashes

8. Software Design

Software Design is one of the most important phases in the development of any software system. It describes how the system will be structured, how different modules will interact with each other, and how the complete application will work internally. For the **Hotel & Restaurant Management System (HRMS)**, the software design is planned in a systematic way to ensure the application is **scalable, secure, maintainable, and easy to use**. The design focuses on creating a web-based platform that can manage hotel room booking and restaurant operations in a single integrated system.

The proposed HRMS is developed using **Node.js and Express.js** for backend processing, **MongoDB** for database storage, and **EJS templates** for creating dynamic user interfaces. The system follows a **modular architecture** where every feature is separated into different components such as authentication, room management, booking management, restaurant menu, orders, billing, and reports. This modular design helps in easy debugging, future upgrades, and better performance.

8.1 System Architecture

The HRMS system is designed using a **three-tier architecture**, which divides the application into three main layers:

1. **Presentation Layer (UI Layer):**
This layer includes EJS pages, HTML, CSS, and JavaScript. It is responsible for displaying forms, dashboards, booking pages, order pages, and reports to users.
2. **Application Layer (Business Logic Layer):**
This layer is handled using Node.js and Express.js. It contains all the logic for booking workflow, room availability, order calculation, invoice generation, and validations.
3. **Data Layer (Database Layer):**
This layer is MongoDB, which stores all the data such as users, rooms, customers, bookings, menu items, orders, and invoices.

8.2 MVC Design Pattern

The HRMS follows the **MVC (Model-View-Controller)** design pattern to organize code properly:

- **Model:** Represents database schemas and handles data operations using MongoDB and Mongoose.
Example: User Model, Room Model, Booking Model, Order Model, Invoice Model.
- **View:** Represents the user interface using EJS templates.
Example: login.ejs, dashboard.ejs, rooms.ejs, bookings.ejs, orders.ejs.
- **Controller:** Contains the logic for handling requests and responses.
Example: bookingController, roomController, orderController

8.3 Authentication and Security Design

Security is a major part of HRMS because it stores customer details and billing records. The system includes:

- **Login/Logout system** for users
- **Role-Based Access Control (RBAC)** for Admin, Receptionist, and Restaurant Staff
- **Password hashing using bcrypt** to protect user credentials
- **Session-based authentication** to keep users logged in securely
- **Input validation** to prevent invalid data entry

8.4 Module-Wise Design

The software design is divided into modules for better workflow:

1. **User Management Module:**
Handles staff registration, login, roles, and permissions.
2. **Room Management Module:**
Admin can add/update/delete rooms, set room price, and manage room status.

3. **Booking Management Module:**
Receptionist can create bookings, allocate rooms, manage check-in/check-out, and calculate stay charges.
4. **Restaurant Menu Management Module:**
Restaurant staff can manage categories and menu items, update prices, and item availability.
5. **Order Management Module:**
Orders can be created for dine-in/takeaway, items can be added, and order status can be updated.
6. **Billing & Invoice Module:**
Generates invoices for hotel booking and restaurant orders with automatic total calculation and taxes.
7. **Reports Module:**
Provides daily sales report, room occupancy report, and revenue summary for management.

8.5 Error Handling and Validation Design

To avoid system crashes and wrong data entry, the system includes:

- Proper form validations (required fields, numeric checks, date checks)
- Error messages for invalid operations
- Middleware-based error handling in Express.js
- Try-catch blocks in controllers for safe execution

9. Database Design

The database is designed using MongoDB collections with relationships managed using ObjectId references.

Collection	Important Fields
users	name, email, passwordHash, role, createdAt
rooms	roomNo, type, pricePerNight, status
customers	name, phone, idProof, address
bookings	customerId, roomId, checkIn, checkOut, status, totalAmount
menuCategories	name, isActive
menuItems	categoryId, name, price, isAvailable
tables	tableNo, capacity, status
orders	tableId, items[{menuItemId, qty, price}], status, total
invoices	bookingId(optional), orderId(optional), grandTotal, tax, createdAt

10. Module Description

Authentication & Roles: Login/Logout, session management, role-based access control for Admin, Receptionist and Restaurant Staff.

Room Management: Add, update, list rooms, set pricing and status (available/occupied/maintenance).

Booking Management: Create booking, manage check-in/check-out, calculate total bill based on stay duration.

Menu Management: Manage restaurant categories and items with price and availability.

Order Management: Create orders (dine-in/takeaway), update kitchen workflow (pending/cooking/served/paid).

Billing & Invoices: Generate invoice for hotel booking and restaurant order, apply taxes and service charges.

Reports: Daily sales report, occupancy report, popular menu items report.

11. Testing

Testing is an essential phase in software development that ensures the application works correctly, efficiently, and securely according to the given requirements. The main purpose of testing is to identify bugs, errors, and missing functionalities before the system is deployed for real use. In the **Hotel & Restaurant Management System (HRMS)**, testing is performed on every module such as authentication, room booking, restaurant order management, billing, and reporting to ensure that the system provides accurate results and smooth performance. Proper testing increases system reliability, improves user experience, and reduces the risk of failure during real-time operations.

The HRMS application is tested using different testing techniques such as **Unit Testing, Integration Testing, Functional Testing, and User Interface Testing**. Each type of testing plays an important role in verifying the correctness of the system.

11.1 Types of Testing Used

1) Unit Testing

Unit testing is performed to test individual components or functions of the system. In HRMS, unit testing is used to check the correctness of modules like login validation, room availability logic, booking calculations, and invoice generation. Each controller function and database operation is tested separately to ensure it works properly without dependency on other modules.

2) Integration Testing

Integration testing is done to verify that multiple modules work correctly together. For example, when a receptionist creates a booking, the room status should automatically change from **Available** to **Occupied** after check-in. Similarly, restaurant orders should connect properly with menu items and billing. This testing ensures smooth communication between routes, controllers, and MongoDB database.

3) Functional Testing

Functional testing checks whether the system features work according to the expected output. This includes testing of room booking workflow, restaurant order placement, invoice generation, and report creation. It verifies that every function performs the correct task and produces accurate results.

4) User Interface (UI) Testing

UI testing ensures that the user interface is simple, responsive, and error-free. Forms, buttons, navigation menus, and dashboards are tested to confirm that they display correctly and are easy to use. Validation messages are also checked to ensure the system guides the user properly when incorrect input is provided.

5) Database Testing

Database testing is performed to verify that data is stored, updated, and retrieved correctly from MongoDB. It ensures that records like customers, rooms, bookings, menu items, and invoices are saved without duplication and maintain proper relationships.

11.2 Testing Strategy

The testing strategy for HRMS includes:

- Testing each module individually before integration
- Verifying all CRUD operations (Create, Read, Update, Delete)
- Validating user input fields to prevent incorrect data
- Checking security and role-based access for different users
- Ensuring correct calculation of booking bills and restaurant order totals
- Verifying error handling and system stability

11.3 Sample Test Cases

Below are some sample test cases performed for HRMS:

Test Case	Description	Expected Result
Login Test	Enter valid username and password	User should login successfully
Invalid Login	Enter wrong credentials	System should show error message
Room Add	Add new room details	Room should be added in database
Room Booking	Book room for customer	Booking record should be created
Check-In	Perform check-in for booking	Room status should change to Occupied
Check-Out	Perform check-out	Room status should change to Available
Order Placement	Add menu items to order	Order total should calculate correctly
Invoice Generation	Generate bill for booking/order	Invoice should show correct total amount
Role Access	Receptionist tries admin feature	Access should be denied

11.4 Result of Testing

After performing all test cases, the HRMS system was found to be working successfully. All modules such as authentication, booking management, restaurant order management, and billing were tested and verified. Errors found during testing were corrected, and the system was improved to ensure stable performance. Testing confirmed that the system produces accurate results, handles invalid inputs properly, and maintains secure access control.

12. Deployment

Deployment is the final stage of the software development process where the completed application is installed and made available for real-time use. The **Hotel & Restaurant Management System (HRMS)** is a web-based application developed using **Node.js, Express.js, EJS, and MongoDB**, and it can be deployed in both **local (on-premise)** and **cloud-based** environments depending on the hotel's requirements. Proper deployment ensures that the system runs smoothly, data is stored securely, and all modules such as booking management, restaurant orders, billing, and reporting are accessible to authorized users.

For local deployment, the HRMS application can be installed on a hotel's computer system or server machine. First, the system requires installation of **Node.js (LTS version)** and **MongoDB** on the server. After that, the project source code is placed in a specific folder and all required dependencies are installed using the command **npm install**. The MongoDB database is configured either on the same machine or a separate local database server. Once the configuration is completed, the application can be started using **npm start** or **node server.js**, and it becomes accessible through a web browser using the server's IP address and port number. This method is suitable for small hotels because it works even without continuous internet access and provides full control over data storage.

For cloud deployment, the HRMS can be hosted on platforms such as **Render, Railway, or any Node.js supported hosting provider**, while the database can be deployed using **MongoDB Atlas (cloud database)**. In this approach, the application code is uploaded to a cloud server and environment variables such as database connection string, session secret, and port number are configured securely. Cloud deployment allows the system to be accessed from multiple devices and locations, making it useful for hotels that require remote monitoring and multi-branch accessibility. It also improves availability and scalability because the server resources can be increased based on system usage.

During deployment, security measures are also considered to protect the system from unauthorized access. The HRMS uses **session-based authentication, password hashing, and role-based access control**, ensuring that only valid staff members can login and access their assigned modules. Proper database backup and recovery strategies are maintained to prevent loss of important customer and billing information. Regular backups can be scheduled automatically in MongoDB Atlas or manually in local deployment.

After deployment, the system is tested again in the live environment to ensure that all modules work correctly, including room booking, check-in/check-out, restaurant order processing, invoice generation, and report generation. Once successful, the HRMS is ready for real-world use and can help hotels and restaurants manage operations efficiently with improved speed, accuracy, and customer satisfaction.

13. Future Enhancements

Future Enhancements describe the additional features and improvements that can be added to the system to make it more advanced, efficient, and useful in real-world scenarios. The **Hotel & Restaurant Management System (HRMS)** is designed with a modular structure, which makes it easy to upgrade and expand in the future. Although the current system successfully manages hotel bookings, restaurant orders, billing, and reports, there are many improvements that can be implemented to increase performance, security, and customer satisfaction.

One of the major future enhancements is the **online payment gateway integration**. Currently, billing is generated inside the system, but in future the system can support online payment options such as UPI, debit/credit cards, net banking, and wallet payments. This will allow customers to pay easily and reduce manual cash handling at the hotel and restaurant. Another important enhancement is the **PDF invoice download and printing feature**, where users can download a professional invoice format directly from the system and share it with customers through email or print it instantly.

The system can also be improved by adding **SMS and Email notification services**. With this feature, customers can receive booking confirmation messages, check-in reminders, and payment receipts automatically. This will improve communication and customer experience. In addition, a **real-time room availability and online booking module** can be added so that customers can book rooms directly from the hotel website without visiting the reception physically. This will make the system more modern and competitive.

Another useful enhancement is the introduction of an **advanced analytics dashboard**. The current reports provide basic information, but in future the system can include graphical reports such as monthly revenue charts, occupancy trends, most ordered menu items, staff performance tracking, and customer feedback analysis. This will help management take better business decisions. The HRMS can also be expanded with a **multi-branch support feature**, where a single system can manage multiple hotel branches from one admin panel. This is useful for large hotel chains.

Additionally, the system can include **inventory and stock management** for the restaurant section. This feature will track raw materials, item stock levels, and automatic low-stock alerts, which will reduce wastage and improve restaurant efficiency. The system can also be enhanced with **mobile-friendly and app-based access**, allowing staff to manage bookings and orders through smartphones or tablets for faster service.

In the future, the HRMS can be made more secure by adding **two-factor authentication (2FA)**, improved user activity logs, and better access control policies. These enhancements will strengthen system security and prevent unauthorized access. Overall, these future upgrades will make the HRMS more powerful, user-friendly, and suitable for large-scale hotel and restaurant management.

14. Conclusion

The **Hotel & Restaurant Management System (HRMS)** is a complete web-based solution developed to simplify and automate the daily operations of hotel booking and restaurant management. In traditional systems, managing room reservations, customer records, restaurant orders, and billing manually can create many problems such as data duplication, calculation mistakes, delays in service, and difficulty in maintaining records. This project successfully solves these issues by providing a centralized and digital platform where all hotel and restaurant activities can be managed efficiently.

The HRMS system provides role-based access for different staff members such as **Admin, Receptionist, and Restaurant Staff**, ensuring secure and organized operations. The Admin can manage overall system settings, rooms, menu items, and reports, while the receptionist can handle customer details, room booking, check-in/check-out, and hotel billing. Similarly, restaurant staff can manage menu categories, orders, table status, and restaurant billing. This division of responsibilities improves workflow efficiency and reduces confusion in day-to-day tasks.

The system is developed using modern technologies such as **Node.js, Express.js, MongoDB, and EJS**, which makes it fast, reliable, and scalable. It supports major features like room management, booking management, restaurant order processing, invoice generation, and reporting dashboards. By storing all data in a database, the system provides quick access to customer history, booking records, sales information, and invoices. This reduces paperwork and ensures that data remains safe and easily retrievable whenever required.

Overall, the **Hotel & Restaurant Management System (HRMS)** achieves the main objective of improving service quality, reducing manual effort, and providing accurate billing and reporting. It is a user-friendly, cost-effective, and efficient solution for small to medium hotels and restaurants. With future enhancements such as online payment integration, PDF invoice generation, SMS/email notifications, and advanced analytics, the system can become even more powerful and suitable for large-scale business use. Hence, HRMS is a successful project that provides a modern approach to hotel and restaurant management with better accuracy, speed, and customer satisfaction.

15. References

The following references were used for studying technologies, designing the system, and implementing the **Hotel & Restaurant Management System (HRMS)** project:

1. **Node.js Official Documentation** – Node.js Foundation
2. **Express.js Official Documentation** – Express Framework Guide
3. **MongoDB Official Documentation** – MongoDB Database Manual
4. **Mongoose Documentation** – ODM Library for MongoDB
5. **EJS Template Engine Documentation** – Embedded JavaScript Templates
6. **HTML5 and CSS3 Documentation** – Web Development Standards
7. **JavaScript Documentation (MDN Web Docs)** – Mozilla Developer Network
8. **Git and GitHub Documentation** – Version Control and Repository Management
9. **Postman Documentation** – API Testing and Development Tool
10. **Software Engineering Concepts** – Pressman, R. S., “Software Engineering: A Practitioner’s Approach”